Experimental investigation on the influence of turbulence on sediment transport

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Overview

1. Introduction
2. Experimental set-up
3. Results
4. Proof of concept
5. Conclusion

Introduction

Background:

- *"I believe that disentangling how near-bed turbulence and bed structure affect bedload transport remains the single most significant challenge in the field"* (Ancey, 2020)
- Phenomenological investigation at the particle scale (Diplas et al., 2008)
- Here, we study the problem at the channel scale (Sumer et al., 2003)
- In particular, we study the effect of turbulence on primary variables, such as concentration and velocity: $\overline{\langle q_s \rangle} = d\overline{\langle C \rangle \langle u_s \rangle}$ (Ballio et al., 2018)

Introduction

Proof of concept:

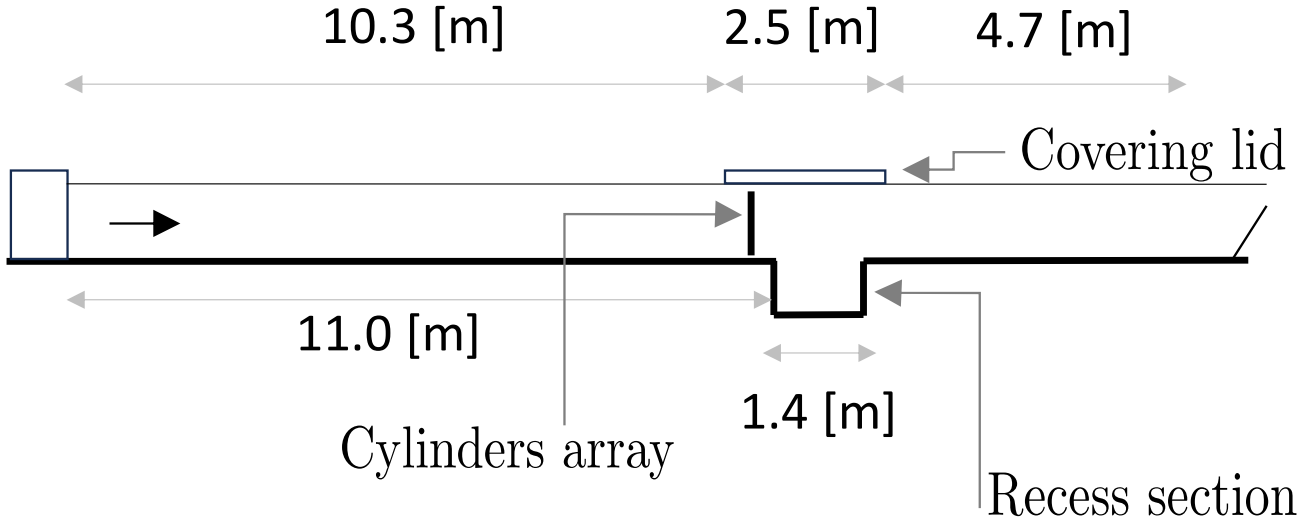
We will compare two possible approaches to describe sediment flowrate:

$$\textbf{Model A: } q_s = f(\tau)$$

$$\textbf{Model B: } q_s = f(\tau, turbulence)$$

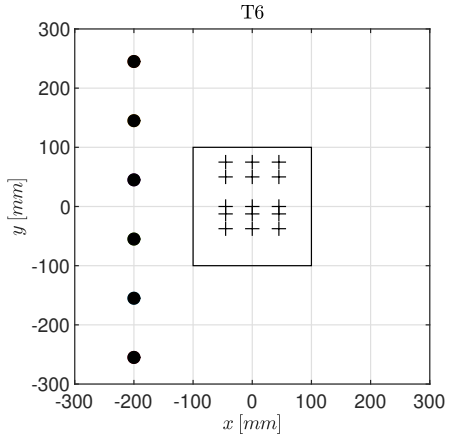
If Model B has less dispersion with respect to Model A, we will conclude that there is an effect of turbulence on sediment transport

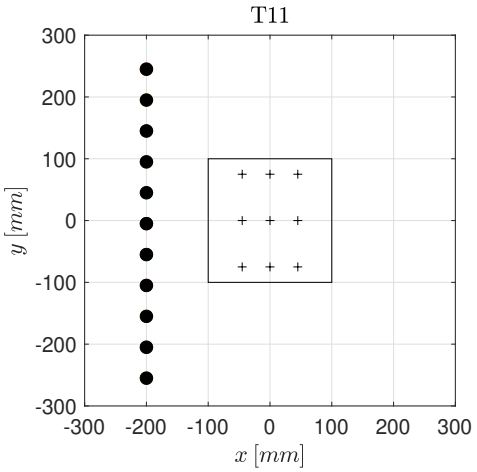
General description

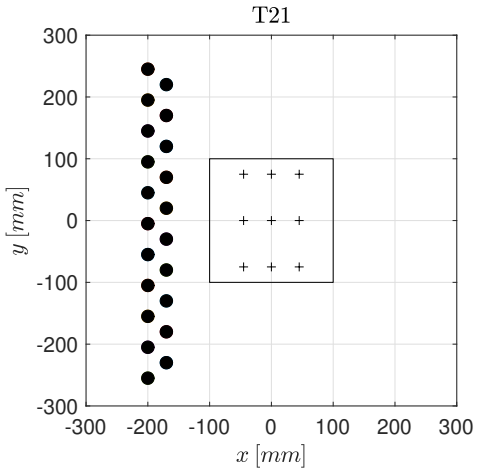


- Berliner flume at Technische Universität Braunschweig
- Gravel (median size = 4 mm) was glued on a series of plates for a length of 17.5 m.
- Between 10.3 and 12.8 m from the inlet, water flowed below a covering lid.
- Different set-ups in the recess section, from 11.3 to 12.7 m.

Introduction







Boundary conditions:

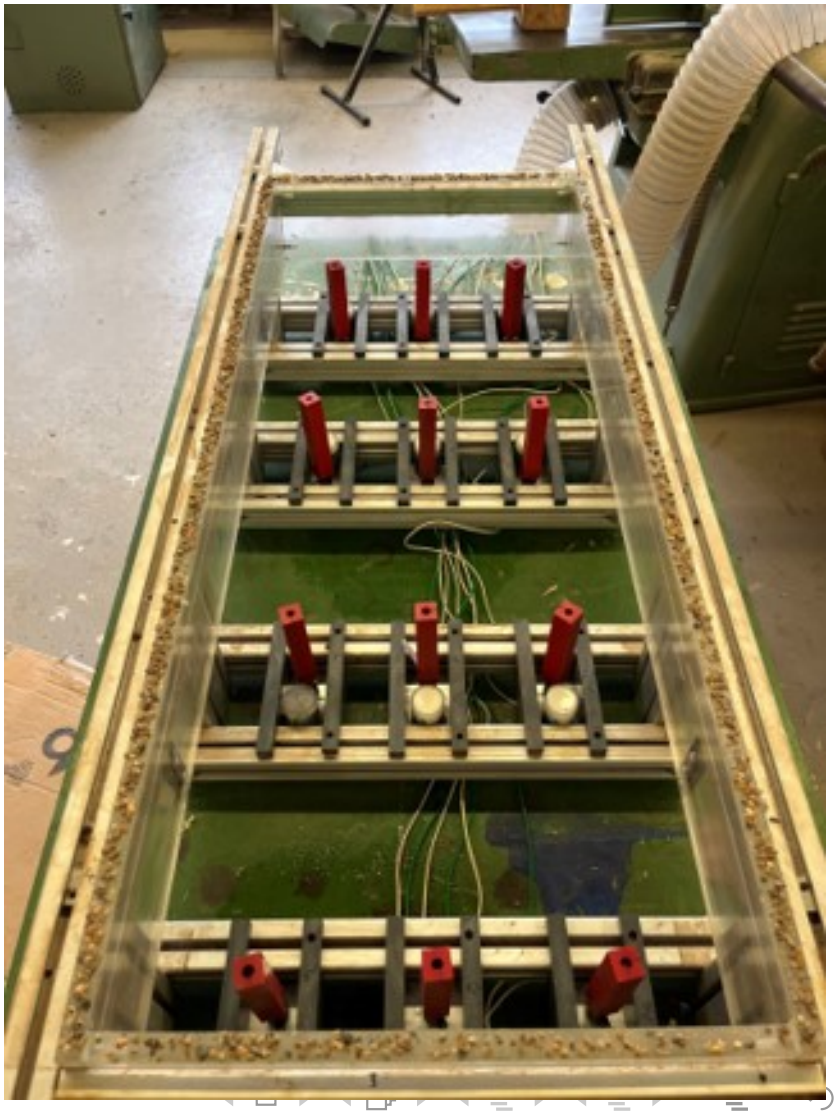
- Externally modified turbulence boundary condition
- Hydraulic boundary condition

Asynchronous measurements:

Bedload, turbulence, and shear stress.

Bed-load set-up

- **Lifing mechanism**.
  The electrical motor (white) moves a threaded rod positioned inside a sleeve (red). Approx 2 mm per impulse.
- **Bed elevation**.
  4 ultrasonic sensors
- **Sediment transport**.
  PTV based on image of difference.



Turbulence set-up

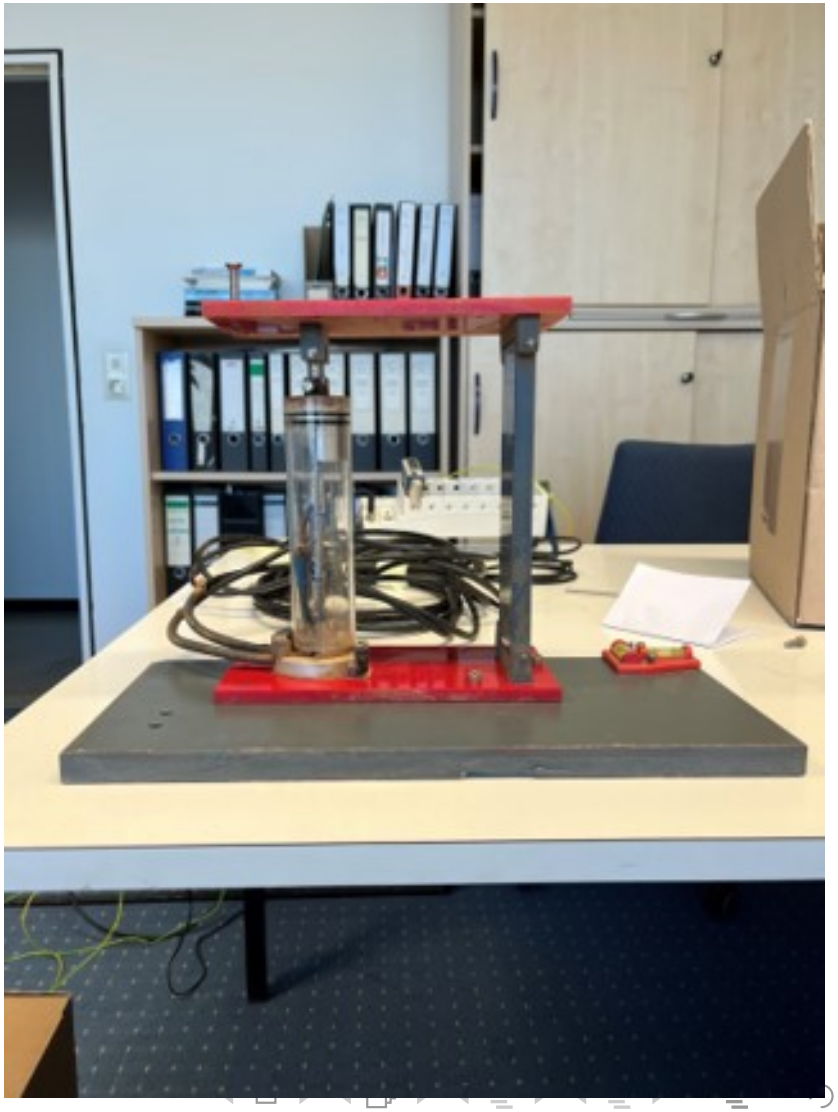
Laser Doppler Anemometry

- Porous bed to ensure similarity w.r.t. the erodible bed
- Vertical grid: 10 pts
- Horizontal grid: 9-15 pts
- Duration: 90 sec or 20k values

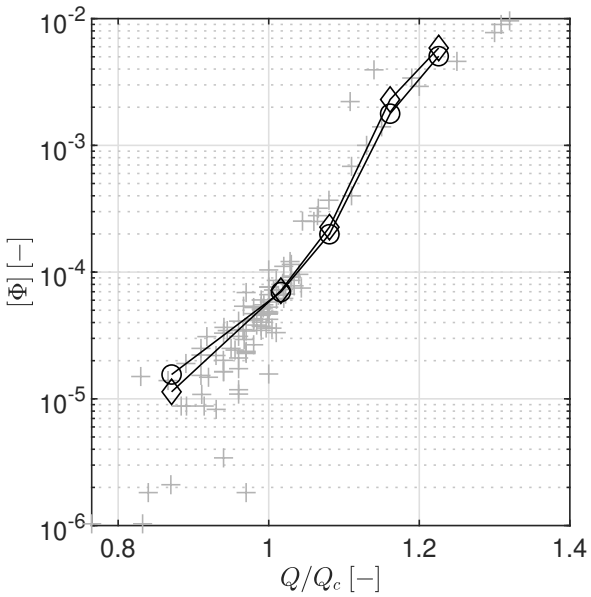
Shear stress set-up

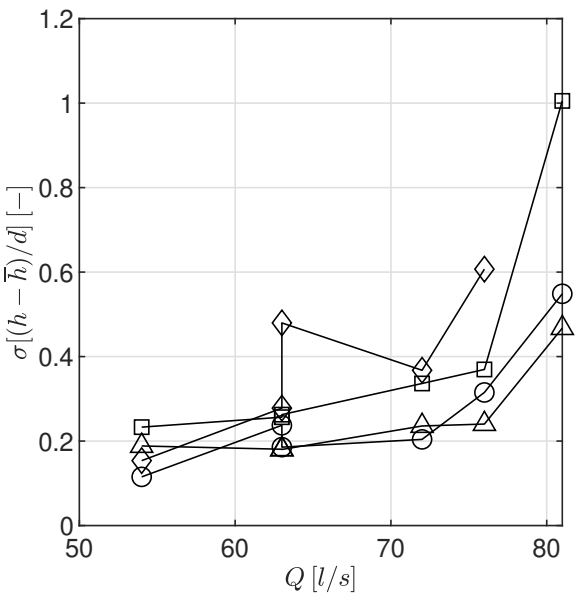
Shear Plate

We measure the deformation of a cantilever beam through two extensimeters at 1615 Hz, then we can estimate the shear force.



Bedload





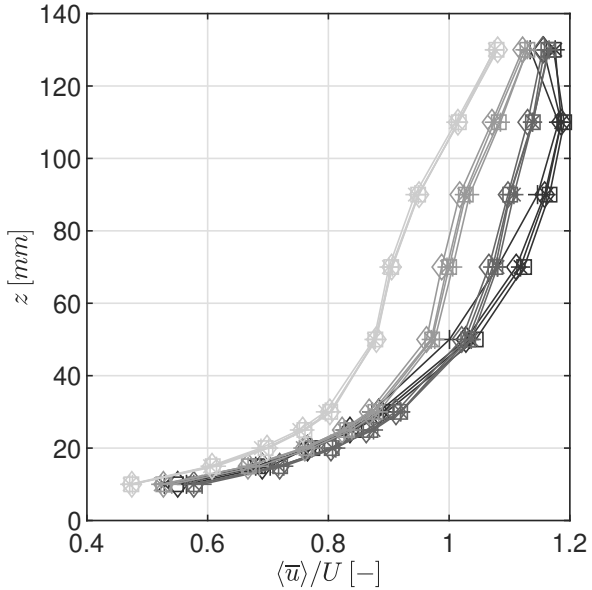
The flowrate is critical $Q = Q_c$ when $\Phi = 6 \cdot 10^{-5}$.

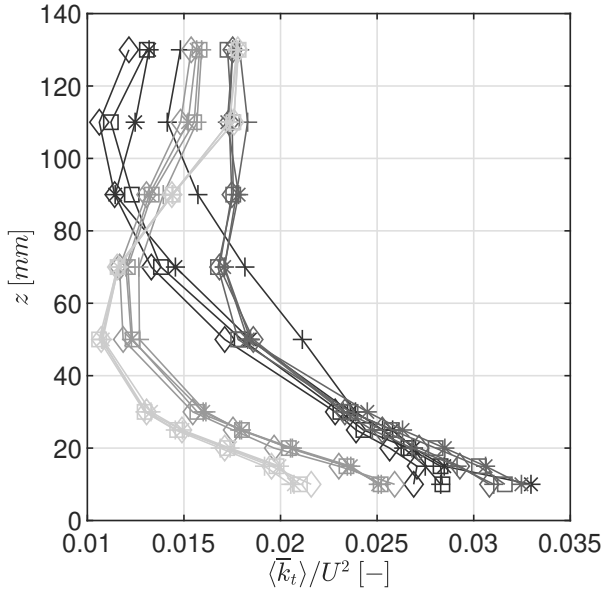
Bed fluctuations: $(h - \overline{h})/d$
Legend: T0 $\circ$, T6 $\diamondsuit$, T11 $\square$, T21 $\triangle$.

$$[q]_{MAN} = N_Q w / T \Delta y \; (\diamondsuit)$$

$$[q]_{PTV} = \overline{(N_A w / \Delta x \Delta y) \langle u_s \rangle} \; (\circ)$$

Turbulence



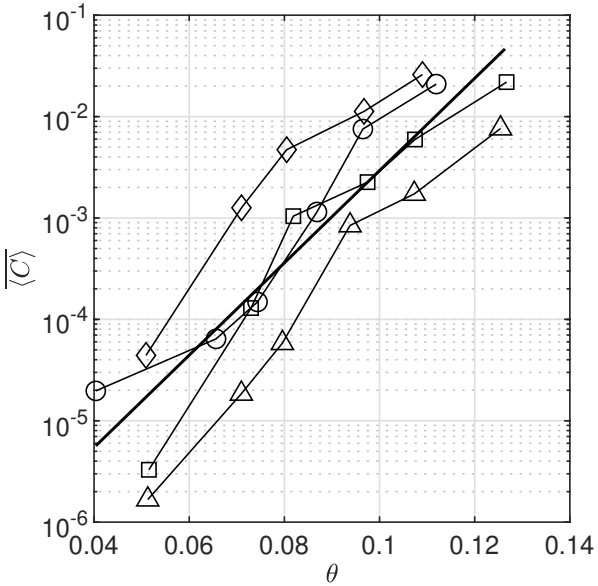


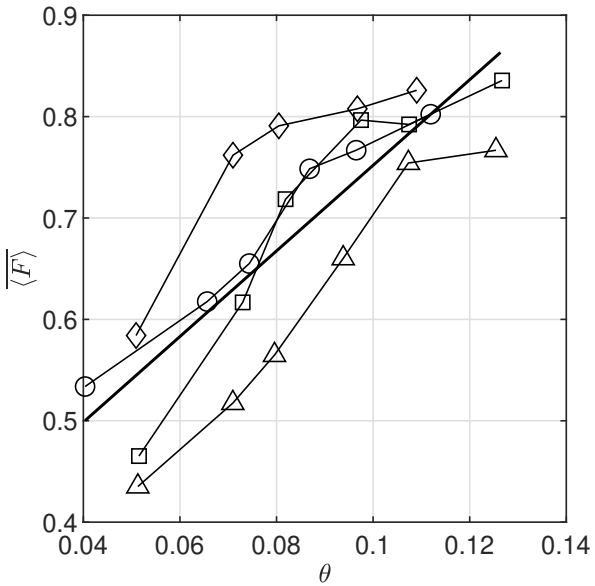
Legend: T0, T6, T11, T21

We will focus on turbulent variables in the log-layer, $z/H \in [0, 0.2]$.

$$k_t^{log} = \frac{5}{H} \int_0^{H/5} \langle \overline{k_t} \rangle dz$$

Model A (shear stress only)





$$\overline{\langle q_s \rangle} = d\overline{\langle C \rangle}\,\overline{\langle u_{s+} \rangle}$$

$$\overline{\langle \Phi \rangle} = \overline{\langle C \rangle}\,\overline{\langle F \rangle}$$

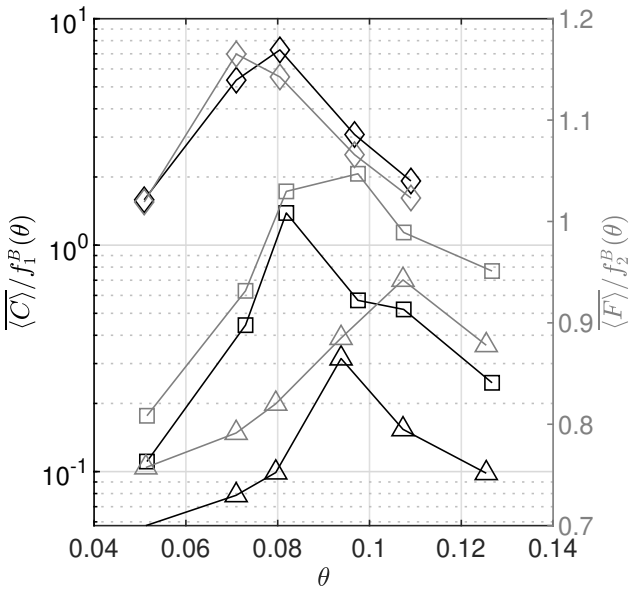
$$\overline{\langle C \rangle} = f_1^A(\theta)$$

$$\overline{\langle F \rangle} = f_2^A(\theta)$$

$$\theta = \frac{\tau}{(\rho_s - \rho)gd}$$

Legend: T0 $\circ$, T6 $\diamondsuit$, T11 $\square$, T21 $\triangle$.

Model B (shear stress and near-bed turbulence)



$$\overline{\langle \Phi \rangle} = \overline{\langle C \rangle}\,\overline{\langle F \rangle}$$

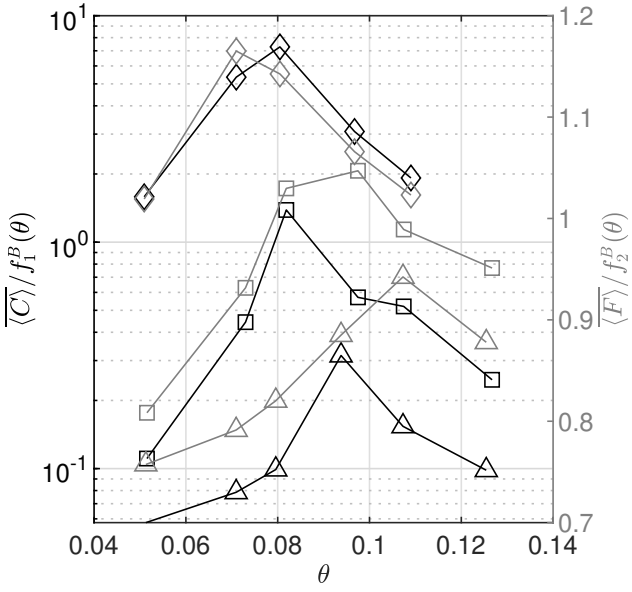
$$\overline{\langle C \rangle} = f_1^B(\theta)$$

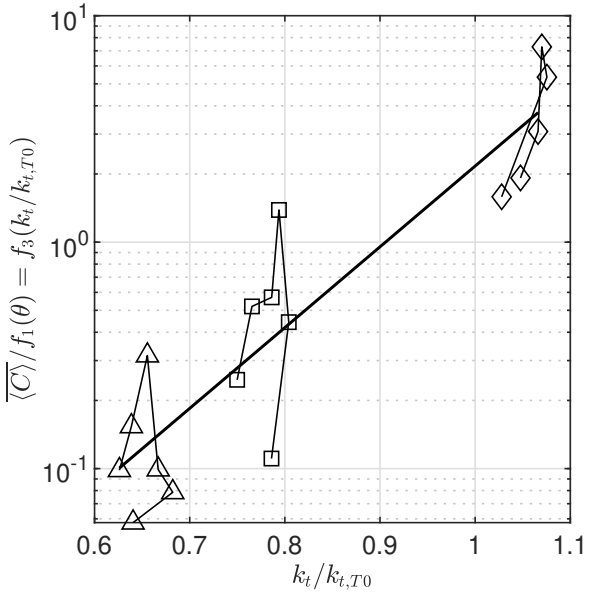
$$\overline{\langle F \rangle} = f_2^B(\theta)$$

Legend: T0 ○, T6 ♢, T11 □, T21 △.

Bare bed data is used for calibrating f_1^B and f_2^B

Model B (shear stress and near-bed turbulence)





$$\overline{\langle \Phi \rangle} = \overline{\langle C \rangle}\,\overline{\langle F \rangle}$$

$$\overline{\langle C \rangle} = f_1^B(\theta)$$

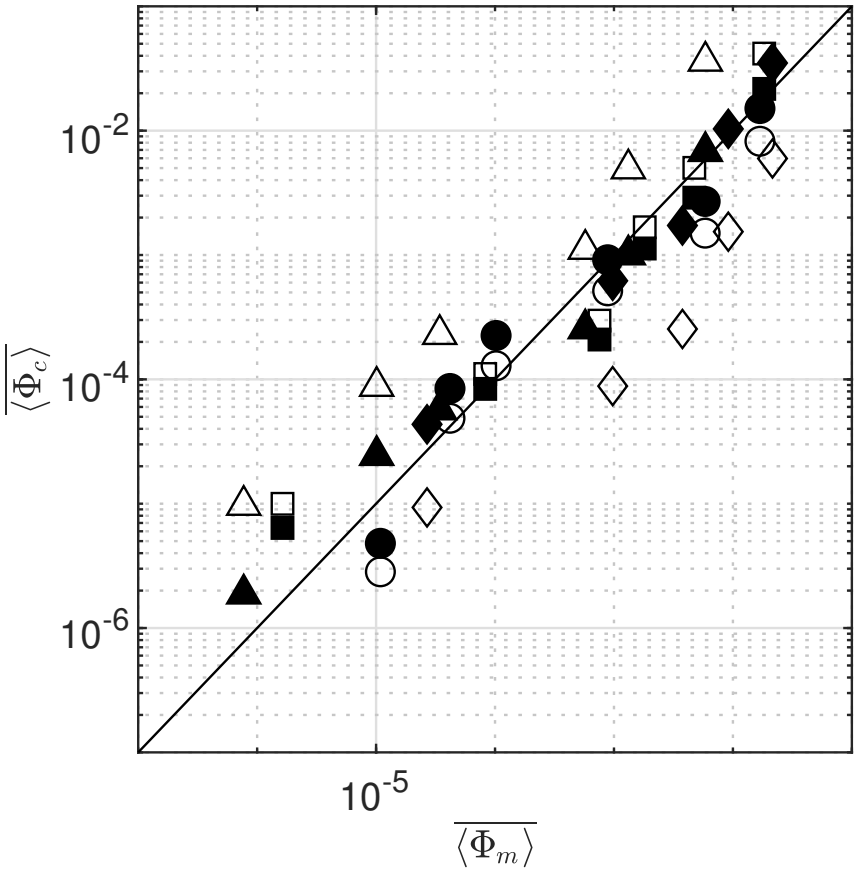
$$\overline{\langle F \rangle} = f_2^B(\theta)$$

Bare bed data is used for calibrating f_1^B and f_2^B

$$\overline{\langle C \rangle} = f_1^B(\theta) f_3^B(k_t^{log}/k_{t,T0}^{log})$$

T6, T11, and T21 data are used for calibrating f_3^B.
Legend: T0 ○, T6 ♢, T11 □, T21 △.

Proof of concept



Legend: T0 ○, T6 ♢, T11 □, T21 △.
Model A: empty marker; Model B: filled marker.

Conclusion

- Sediment flowrate is controlled by average shear stress and turbulent kinetic energy;
- In particular, a change of velocity fluctuations affects concentration more than solid velocity.

I thank **K. Koll** and **J. Aberle** for accepting me in their research group at Leichtweiß-Institut für Wasserbau. We thank **U. Ecklebe** and **C. David** for their help in building the experimental apparatus. Advices from **J. Aliaga** relative to the experiments are highly acknowledged.

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Particle tracking velocimetry

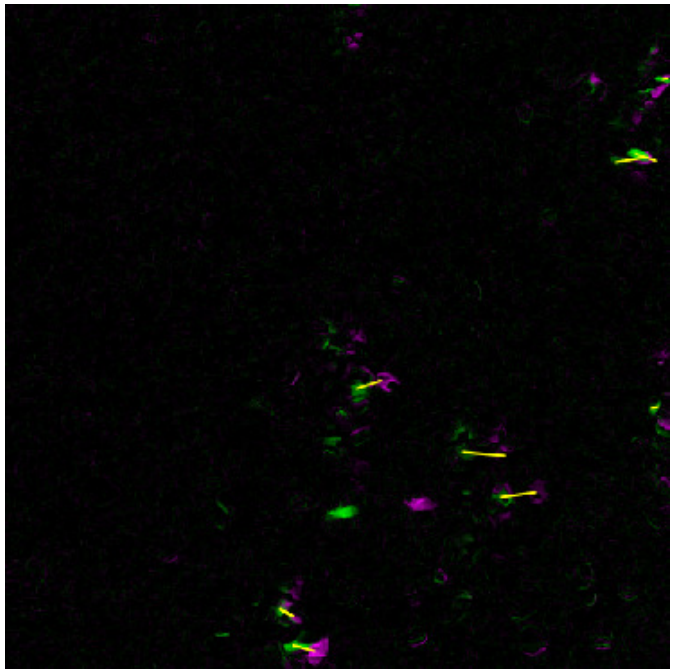
Image Preparation:

- Definition of the homography matrix
- Conversion from RGB to BW
- Subs traction of two consecutive frames
- Homography transformation
- Cropping, to retain only the investigated area

Image of Difference Analysis:

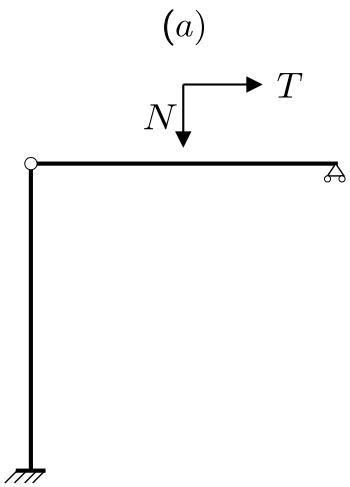
- Spatial filter (camera vibration, light fluctuations)
- Particle identification
  - Minimum intensity of a blob
  - Minimum size of a blob
- Velocimetry based on cost minimization. Here we define the cost as the intensity difference between two consecutive IoD

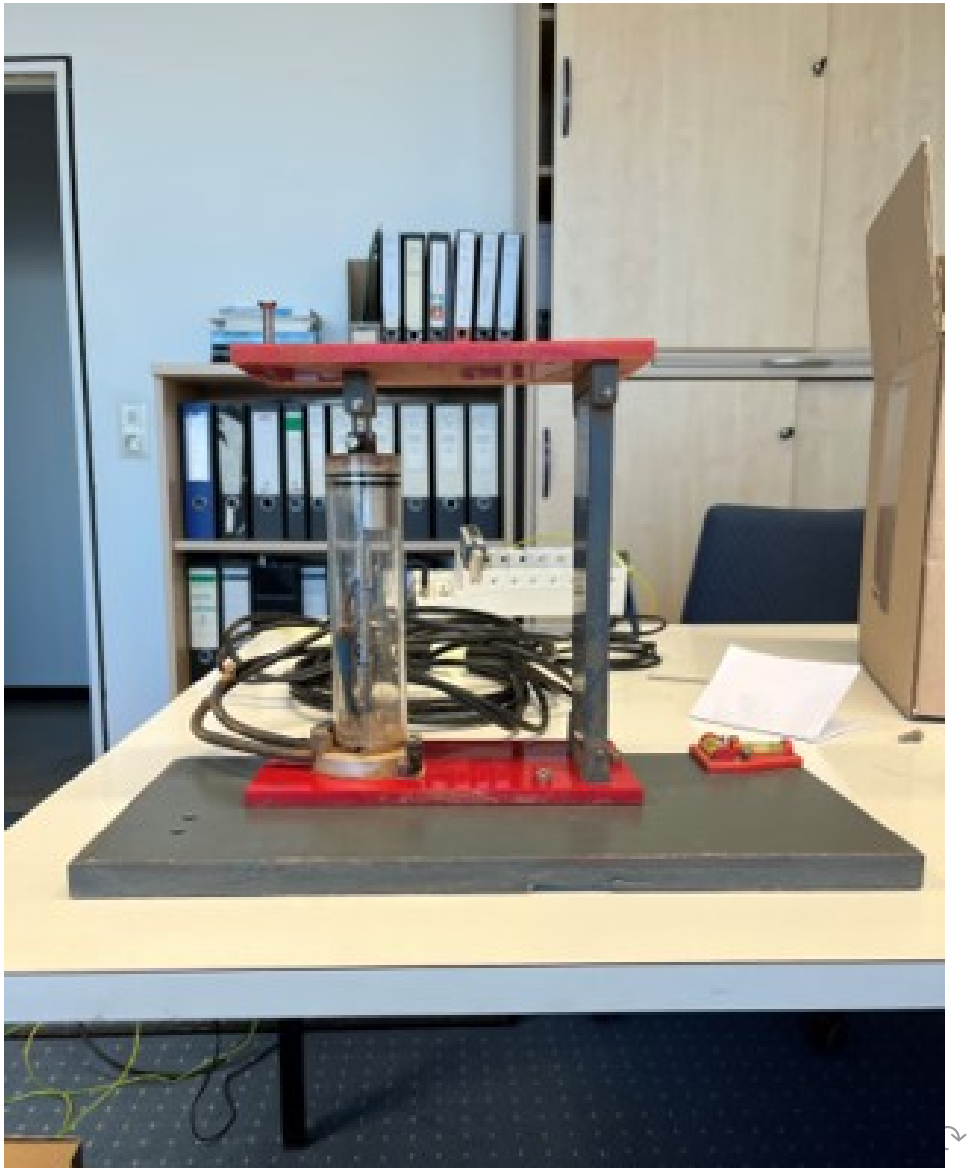
Particle tracking velocimetry



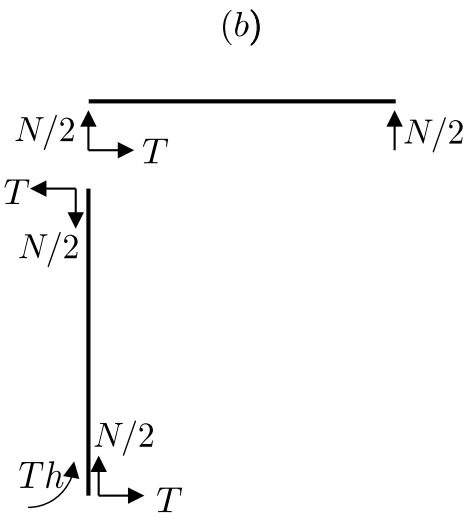
Green blobs: IoD1. Magenta blobs: IoD2. Yellow line: velocity vector.

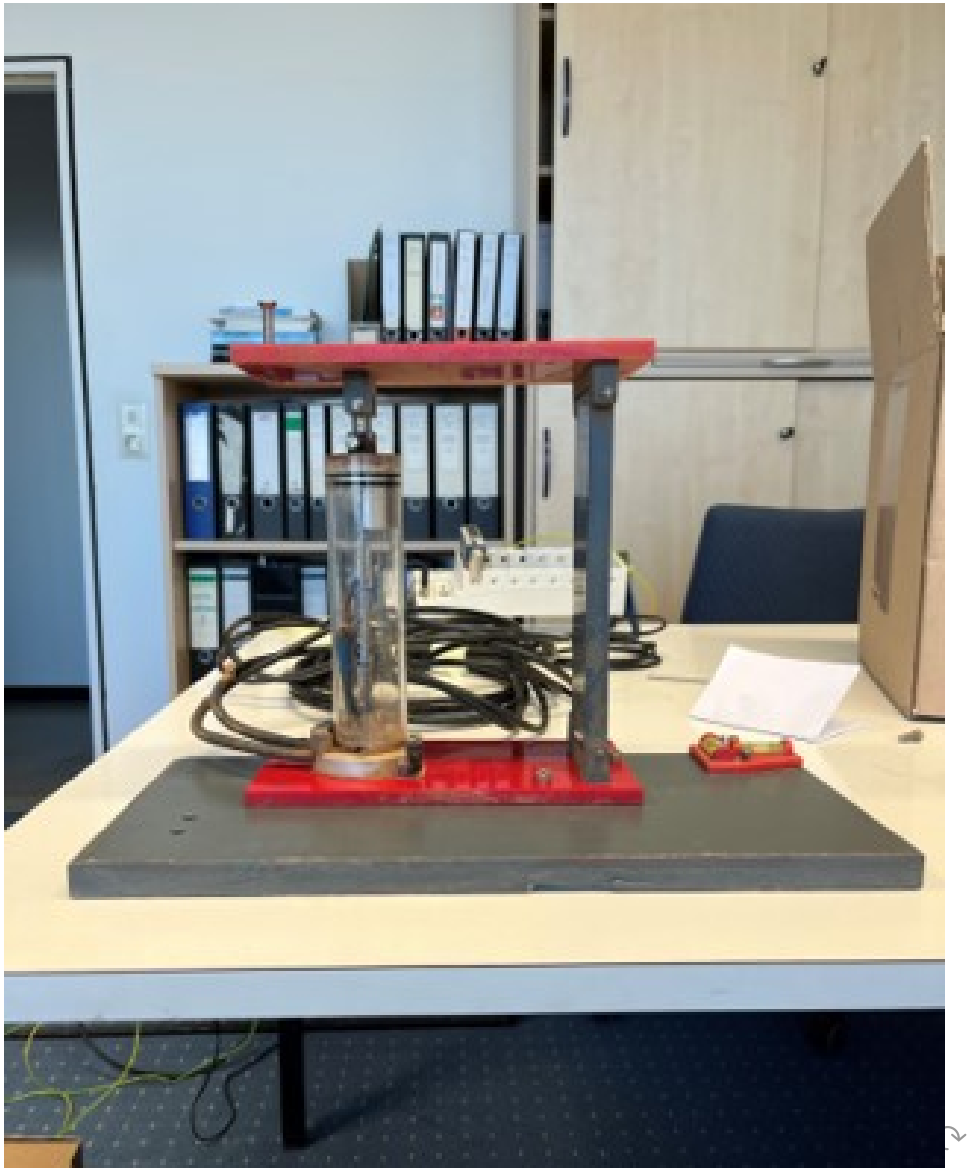
Shear plate



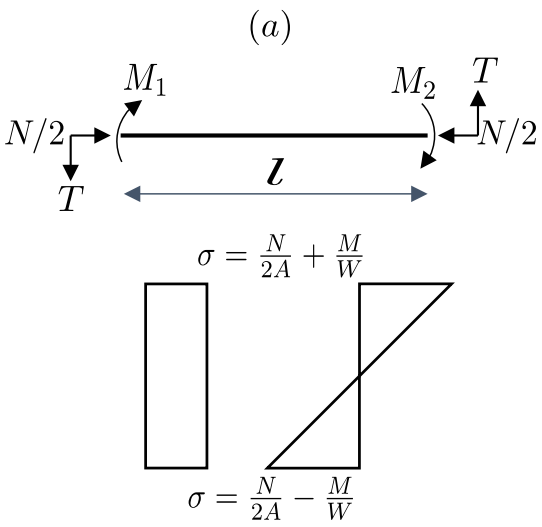


Shear plate



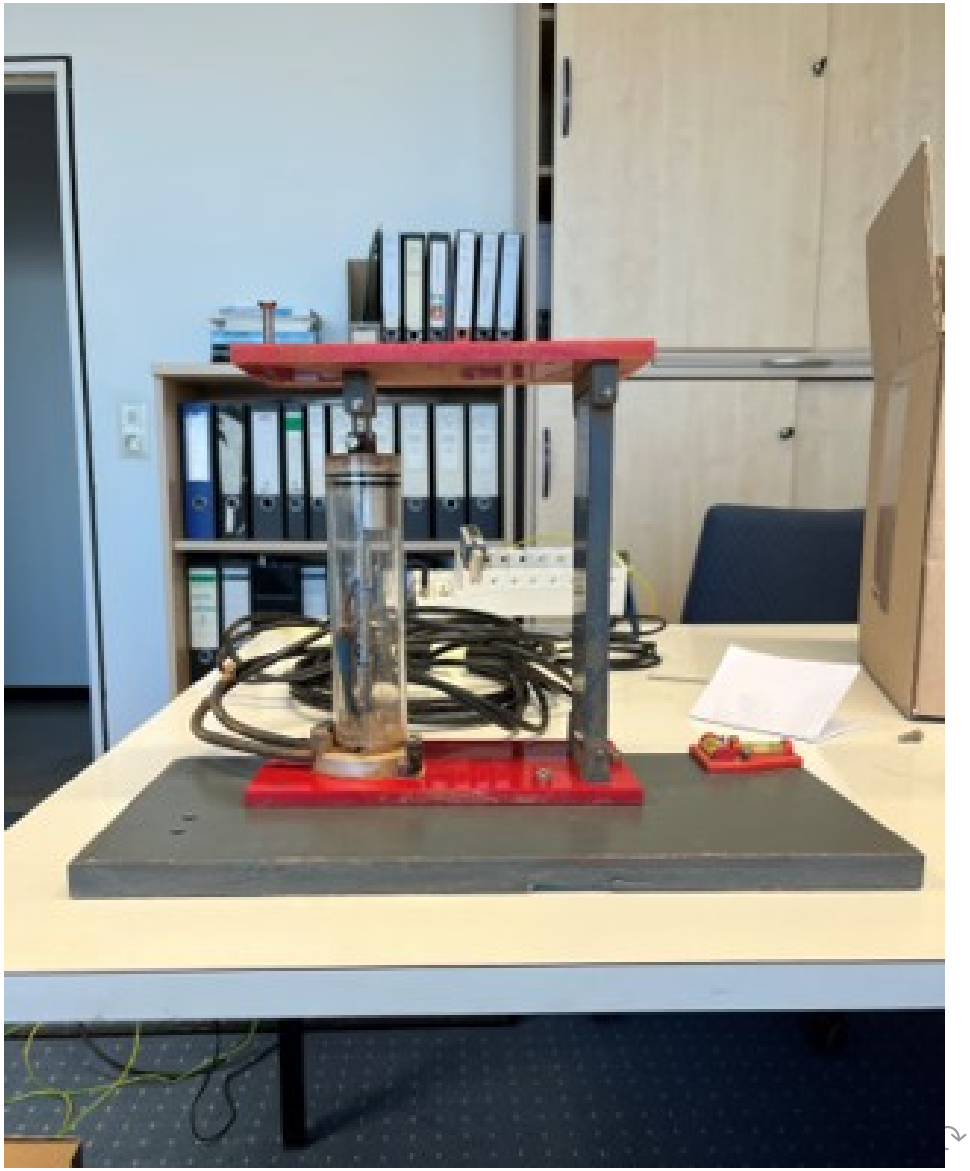


Shear plate



$$T = \frac{(M_1 - M_2)}{l}$$

$$\sigma = \sigma_c + \sigma_b = \frac{N}{2A} + \frac{M}{w}$$

$$\sigma = \varepsilon E$$

$$F = (\varepsilon_{b2} - \varepsilon_{b1}) \frac{EW}{l}$$